

Garrett Kunkler

✉ gnkunkler@gmail.com 🔗 gkunkler.github.io in gkunkler 🌐 gkunkler

Education

California Polytechnic State University, B.S. Mechanical Engineering

San Luis Obispo, CA
Sep 2021 – (June 2026)

- **University Honors Program**
- **GPA:** 4.0/4.0
- **Minors:** Physics, Mathematics
- **Coursework:** Dynamics, Mechanics, Linear Analysis, Thermodynamics, Fluid Mechanics, Modern Physics, Circuits, Quantum Mechanics

KTH Royal Institute of Technology, International Exchange

Stockholm, Sweden
Aug 2024 – June 2025

- **Grade:** A
- **Coursework (Master's Level):** Acoustics, Medical Engineering, Electrodynamics, Game Theory, Computational Fluid Dynamics, Optimization, Environmental Physics

Academic Experience

UC Santa Barbara, Physics Research Experience for Undergraduates (REU)

Santa Barbara, CA
June 2025 - Aug 2025

- Assessed systematic uncertainties in the MicroBooNE neutrino experiment
- Prototyped machine learning model for particle selection

KTH Medical Imaging, Undergraduate Physics Research

Stockholm, Sweden
Aug 2024 - June 2025

- Created module in open-source simulation framework Allpix Squared
- C++ simulation of particle interactions in CT scanning sensors
- Presented at Allpix Squared workshop in Amsterdam ([Slides](#))

VERITAS, Undergraduate Physics Research

San Luis Obispo, CA
Sept 2023 - June 2024

- Analyzed gamma ray spectra from blazars
- Organized and presented preliminary results in student research fair poster
- Co-authored on conference presentation and in-progress paper

Cal Poly SLO, Physics Learning Assistant

San Luis Obispo, CA
Jan 2023 - June 2024

- Facilitated in-class discussion with physics students
- Prepared for lessons with pedagogical discussions with faculty

Industry Experience

Dexcom, Systems Engineering Intern

San Diego, CA
July 2024 – Aug 2024

- Designed physical impact testing for wearable glucose sensors
- Created detailed timeline to ensure coordination between long lead-time tasks
- Modeled and 3D printed test fixtures to fit with existing equipment
- Analyzed and presented data using DOE (Design of Experiment) software

Yaskawa, Engineering Intern

Santa Clara, CA
June 2023 – Sep 2023

- Embedded robot controller programming for data integration
- Kinematics derivations and microservice communications
- 1D displacement laser measurements and hypothesis testing

Projects

Herbie Robotics Project

Jan 2023 - Dec 2023

- Applied for and awarded grant funding
- Built an interactive and educational trailer for an autonomous robot
- Presented to local preschoolers

Personal Coding Projects

gkunkler.github.io/coding/

- Physics Monte Carlo Simulations, Language Parsing, etc.

Cal Poly Robotics Club

Spring 2022 - Spring 2024

- Designing and assembling components of ongoing [astromech project](#)
- Created intentional onboarding projects to improve retention of new members

Certifications

Oracle Certified Associate, Java SE 8 Programmer

Skills

Programming: Python, MATLAB, Java, C++, Typescript/React, C#, IEC 61131, GitHub, LaTeX, Linux, Jupyter Notebooks

Technologies: CAD (Solidworks, Onshape), 3D Printing, JMP